Stat 618 (Bayesian Statistics)

Homework # 1: Bayes Theorem

Due Jan 31. Quiz 1 is on Jan 31

1. (Christensen, exercise 2.1) An enzyme-linked immunosorbent assay (ELISA) test is performed to determine if the human immunodeficiency virus (HIV) is present in the blood of individuals. The ELISA test is not perfect. Suppose that the ELISA test correctly indicates HIV 99% of the time, and that the proportion of the time that it correctly indicates no HIV is 99.5%. Suppose that the prevalence among blood donors is known to be 1/10,000. What proportion of blood that is donated will test positive using the ELISA test? Also, what proportion of the blood that tests negative on the ELISA test is actually infected with HIV? Finally, what is the probability that a positive ELISA outcome is truly positive, that is, what proportion of individuals with positive outcomes are actually infected with HIV?

2. In class, we solved the following exercise.

   There exists a test for a certain viral infection. It is 95% reliable for infected patients and 99% reliable for the healthy ones. That is, if a patient has the virus (event V), the test shows that (event S) with probability $\boldsymbol{P}\{S \mid V\} = 0.95$, and if the patient does not have the virus, the test shows that with probability $\boldsymbol{P}\left\{\overline{S} \mid \overline{V}\right\} = 0.99$. Suppose that 2% of all the patients are infected with the virus, $\boldsymbol{P}\{V\} = 0.02$. If the test shows positive results, what is the (conditional) probability that a patient has the virus?

   The answer was ≈ 0.66. Now, suppose that three independent tests all gave positive results. Given that, what is the (conditional) probability that the patient has the virus?

3. An insurance company divides its customers into 2 groups. Twenty percent of customers are in the high-risk group, and eighty percent are in the low-risk group. The high-risk customers make an average of 1 accident per year while the low-risk customers make an average of 0.1 accidents per year. The number of accidents is modeled by a Poisson distribution.

   (a) Mary had no accidents last year. What is the probability that she is a high-risk driver?

   (b) What is the prior distribution, the posterior distribution, and the marginal probability of no accidents last year?

   (c) Mary continues driving. After three years with the company, she still has no traffic accidents. Now, what is the conditional probability that she is a high-risk driver?

4. On an exam, students X and Y forgot to sign their papers. Professor knows that they can write a good exam with probabilities 0.8 and 0.6, respectively. After the grading, he notices that one unsigned exams is good and one is bad. Given this information, and assuming that students worked independently of each other, what is the probability that the good exam belongs to student X?

Rules of Probability

Probability of a union (disjoint events)	$P(A \cup B) = P(A) + P(B)$
Probability of a union (general case)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Probability of an intersection (independent events)	$P(A \cap B) = P(A) \cdot P(B)$
Probability of a complement	$P(\bar{A}) = 1 - P(A)$
Bayes Rule	$P(A\|B) = \dfrac{P(B\|A)P(A)}{P(B)}$
Total Probability	$P(A) = P(A\|B)P(B) + P(A\|\bar{B})P(\bar{B})$
	$P(A) = \sum_{j=1}^{n} P(A\|B_j)P(B_j)$

Thomas Bayes (1702–1761)